

Koa Shen

B.S. Candidate in Mechanical Engineering · UC Santa Barbara · Robotics · Motion Systems

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EDUCATION

University of California, Santa Barbara

Expected June 2028

B.S. Mechanical Engineering — 3.97 Major GPA · 3.94 Cumulative GPA

Santa Barbara, CA

- Relevant coursework: Statics, Dynamics, Dynamical Systems and Machine Learning (in progress), Mechanics of Materials, Circuits, MATLAB, and Calculus-based Physics.
- Consistently earned Dean's Honors in the College of Engineering.

EXPERIENCE

Mechanical Engineering Intern

Summer 2026 – Present (extended to part-time remote)

SLAC National Accelerator Laboratory — LCLS

Menlo Park, CA

- Built CAD import, kinematics, collision-check, and load-case workflows for high-DOF LCLS assemblies.
- Applied XCS findings to the optics redesign and documented Solid Edge, Teamcenter, and EPICS interfaces.

Maintenance Technician

Summer 2025

Exploratorium

San Francisco, CA

- Serviced 100+ exhibits and led three overhauls; the Arp Forms redesign avoided \$3,000+ in replacement costs.

Fabrication Team Lead

Fall 2024 – Present

Gaucha Racing (FSAE EV) — UC Santa Barbara

Santa Barbara, CA

- Led 15 people through GR26 safety-system design and installation; passed FSAE technical inspection.
- CNC-machined 33 parts from 4130 steel and 7075-T6 aluminum; held fits to 0.0005 in and standardized five spindles.

SELECTED PROJECTS

Digital Twin Simulation Framework for LCLS Motion Stages

2026 – Present

- Built a 32-DOF Drake twin (305 bodies) with OpenCascade, CoACD, MeshCat; millisecond checks span 15 LCLS stages.
- Applied it to three assemblies; retroactive XCS checks exposed at least five pinch points restricting use.

Polycapillary Optics Assembly Redesign

2026 – Present

- Developed a redesign for a 32-DOF optics assembly with three kinematic bases and six hot-swappable optics; projected savings of 1–2 hours per shift-day and roughly \$30k in annual value are not yet verified.

Compact Cycloidal Reducer & Stepper Package

Summer 2026 – Present

- Designed a 15:1 cycloidal reducer for NEMA 17; planned arm targets (unvalidated): 6 DOF, 381 mm reach, 0.5 kg payload.
- Built PlatformIO C++ firmware for TMC2209 stepper control, including step/dir generation and real-time telemetry.
- Integrated a 12-bit AS5600 encoder for missed-step detection; multi-joint output sensing remains planned.

FSAE Steering Wheel Development

Fall 2024 – Present

- Integrated wheel buttons, potentiometers, display, and quick-release; FEA-guided iteration cut mass from 5 lb to 1.7 lb.

TECHNICAL SKILLS

CAD & Manufacturing: SolidWorks, Solid Edge, Teamcenter PDM, Inventor / Fusion, Mastercam, GD&T, CNC / manual machining, Composites, Welding, Sheet metal

Programming: Python, C/C++, PlatformIO, MATLAB, Arduino / microcontrollers, Git / GitHub

Robotics & Controls: Drake kinematics (path planning in development), Collision detection, TMC2209 stepper control, AS5600 encoders / I2C multiplexing